

Exp. No: 2

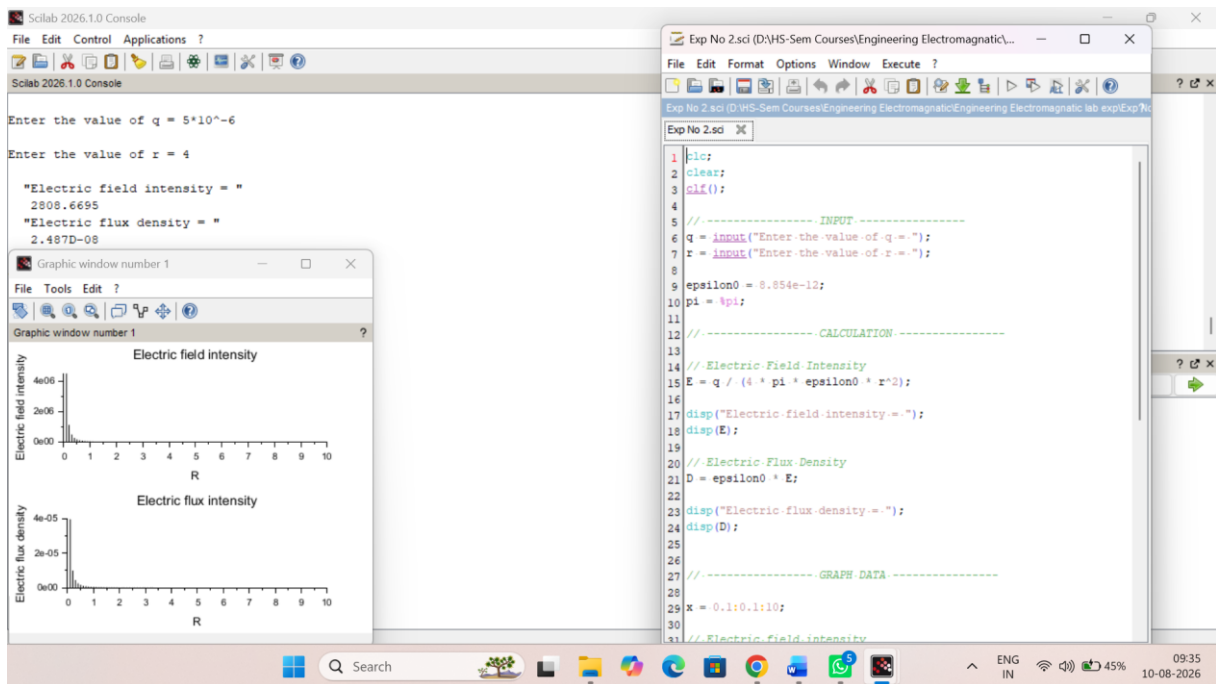
Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

Aim:

To write program for electric field and electric flux density in sphere using SCILAB.

Software required:

- SCILAB version 6.0.



EXP NO :- 02

$$q = 5 \times 10^{-6}$$

$$r = 4 \text{ cm}$$

Electric Field

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

$$\left(\frac{1}{4\pi\epsilon_0} \right) = 9 \times 10^9 \text{ N/C}$$

$$E = \frac{9 \times 10^9 \times 5 \times 10^{-6}}{4}$$

$$= \frac{45 \times 10^3}{16}$$

$$E = 2812.5 \text{ N/C}$$

$$\therefore \underline{281}$$

$$E = 2809.33 \text{ N/C}$$

Electric Flux Density D

$$D = \epsilon_0 E$$

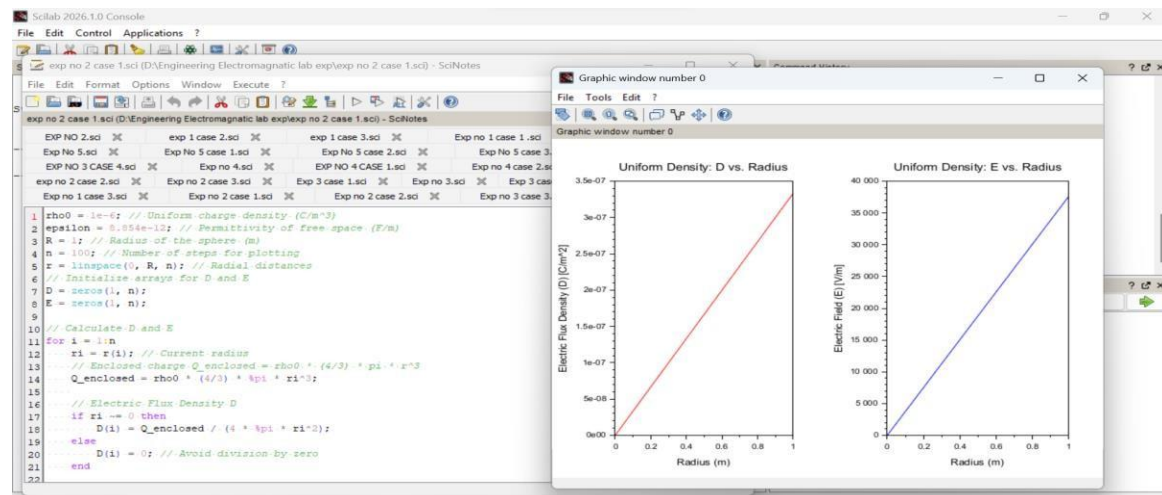
$$D = \frac{q}{4\pi r^2} = \frac{5 \times 10^{-6}}{4\pi (4)^2} = \frac{5 \times 10^{-6}}{201.06}$$

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$$D = 2.49 \times 10^{-8} \text{ C/m}^2$$

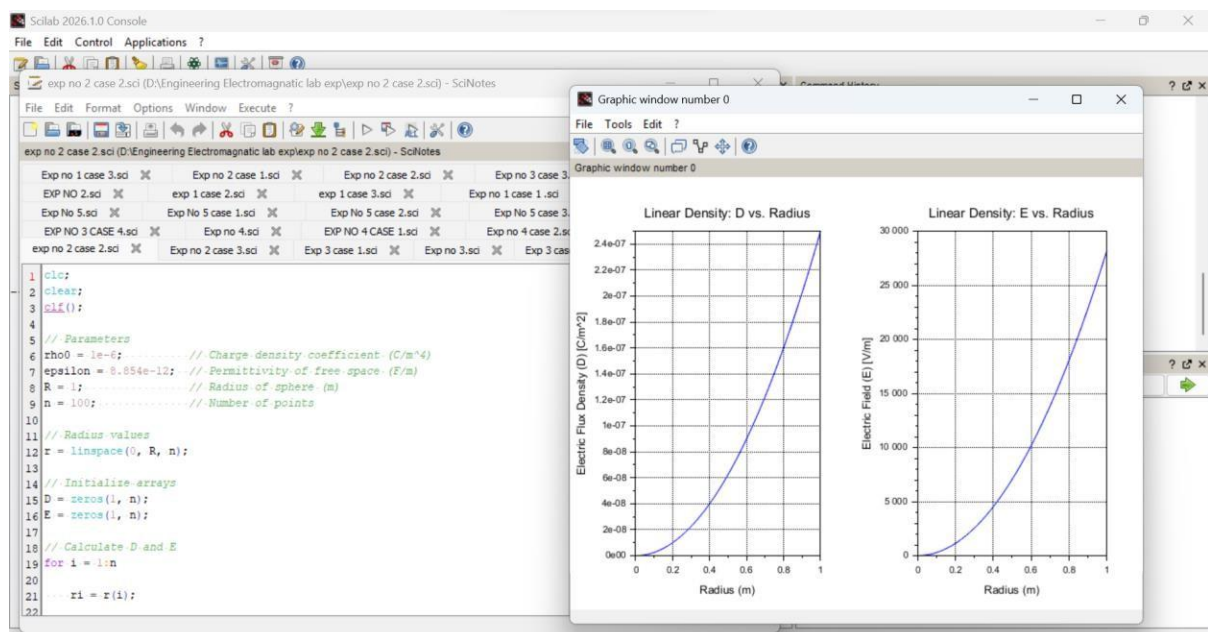
Case 1:

Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ_0 . What are the values of E and D at the center and at the surface of the sphere?



Case 2:

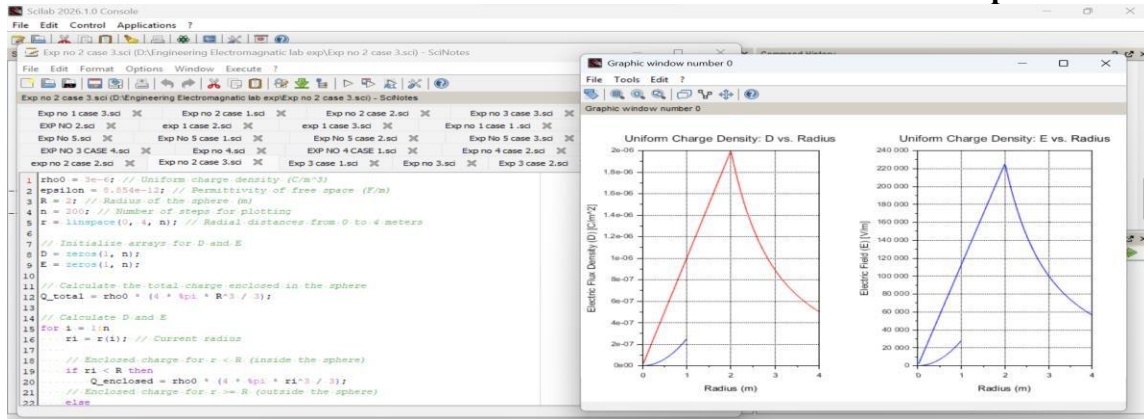
For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive the expressions for the electric field (E) and electric flux density (D). Plot their variations inside the sphere and analyze the trends.



Case 3:

A sphere of radius $R=2$ m has a uniform charge density $\rho=3 \times 10^{-6}$ C/m³

1. Calculate and plot the electric flux density $D(r)$ and electric field $E(r)$ as a function of radial distance r , ranging from 0 to 4 meters.
2. Discuss the behaviour of the electric field inside and outside the sphere.



Result:

Thus, the program was verified for electric field and electric flux density in sphere using SCILAB was successfully.